# Coffee Shop Sales Analysis

## Introduction

This project focuses on sales data from a coffee shop located in different parts of the New York City area. It includes specific details from each transition taken place over the course of 12 months. The data gives us great insights such as timing, product, amount paid and more, allowing us to dive deeper into which item generates the most revenue for this café, and especially how it alters throughout different seasons. This can help us with understanding inventory needs and provide insights on which advertisements to display on the board outside the café according to seasonality.

Our main focus will be to understand which drink or item brings in the most revenue according to seasonality. Café's struggle with advertising the right product at the right time, since consumer taste changes as the seasons do, especially in an area like New York City. A dive into descriptive analysis first, to find most popular items or drinks by season will help with this problem. Predictive analysis will be used on the dataset later to predict which item will bring in the most revenue in the future, according to season, so that the café can adjust it's advertisement and marketing strategies accordingly. Additionally, inventory can be decided based on the model. However, inventory will not be the main focus in this project, **generating and maximizing revenue by most popular seasonal item is the main focus.**

Potential benefits include having a better understanding of customer's taste, and maximizing revenue when adjusting advertising and marketing strategies.

## Objectives

### Main Objective

Perform descriptive analysis on dataset, deriving most popular product by seasonality. After, perform predictive analysis on the dataset, predicating which product will have the most sales according to each season. 2-3 different models will be used to perform the predictive analysis part of the project, ultimately picking the model with the best overall performance, and tuning it to further improve the model.

### Specific Objectives and Milestones

1. Clean data: Data is cleaned for the most part, but it needs to be altered to display seasons across the year, the dates column will be needed for that. Deleting some unnecessary columns like time of transaction will be useful, since we are interested in seasonality not hourly sales revenue.
2. Descriptive analysis: Diving deep into most popular drink per season. Recording that in documents.
3. Correlation Matrix: Looking at which features are highly correlated.
4. Standardize the data.
5. Pick 2-3 models to train the data on.
6. Evaluate performance for these models, and pick the best performing one.
7. *Optional*: tune the model for better performance.

8. Document findings from descriptive analysis, and prescriptive analysis. Such as, model used, accuracy and benefits.
9. Provide marketing strategies based on predictions, and offer advertisement strategies based on descriptive analysis.

## Dataset Description

### Data Source

The data is derived from Kaggle.com, and includes coffee shop transactions over a period of 12 months.

### Data Characteristics

***Features:*** 11 features in the dataset, ranging from numeric, text and datetime. These features include transaction id, transaction date, transaction time, transaction quantity, store id, store location, product id, unit price, product category, product type, and product details.

### Data Preparation

1. Deletion of certain features will be certain, such as the id columns. It is not needed for this type of analysis, thus, store id, product id, and transaction id will be deleted.

2. Product detail column will be split into description of the product sand then size of the project. Since it contains the size and description in one column, this can result in messy data. So, a new column(size) will be created to organize that information.

3. *Time of transaction might be deleted, this may or may not be helpful for analysis.*

4. No missing values recorded in the data.

## Methodology

### Machine Learning Models

Since the predicting outcome will be a category instead of numeric, this project will treated as a classification problem. The three models that will be trained are:

1. Multiclass Logistic Regression
    - Good for understanding which features (price, season, product type) contribute most.
    - Useful because it gives coefficients and clear patterns.

2. Random Forest Classifier
    - Handles non-linear relationships very well.
    - Good with categorical + numeric mixed data.

- Works well with seasonality, product categories, and pricing features.
- Naturally handles interactions

### Evaluation Metrics

- Accuracy – overall correctness, easy for stakeholders to interpret.
- Precision & Recall – important because some products may be minority classes.
- F1 Score – balances precision and recall; ideal when class distribution is uneven.
- Confusion Matrix – helps understand which products are being misclassified.

### Training and Testing

70% training / 30% testing, or 80/20 since dataset is large. Stratify by season to prevent one season dominating the training data.

Cross-Validation: Use k-fold cross-validation (k=5) for more robust model comparison.
Ensures results are not dependent on one random split.

Hyperparameter tuning may be done based on which model performed best, if it is Gradient Boosting model or Random Forest, then some adjustments & tuning may be done such as:

- ***Random Forest:*** *n_estimators, max_depth, min_samples_split*

## Outcomes

The goal of the analysis was to predict product type demand based on transactional and seasonal data. Using two supervised learning models. Logistic Regression and Random Forest, the study achieved strong predictive performance and revealed meaningful business insights.

1. Logistic Regression

- Accuracy: 70.20%
- Precision: 71.03%
- Recall: 70.20%
- F1 Score: 70.04%

Logistic Regression provided a solid baseline model but struggled with the complexity and non-linear relationships of customer purchasing patterns.

2. Random Forest Classifier

- Accuracy: 91.73%
- Precision: 92.17%
- Recall: 91.73%
- F1 Score: 91.79%

Random Forest significantly outperformed Logistic Regression, confirming that non-linear models capture product-season interactions far better. This model was selected as the final predictive model.